

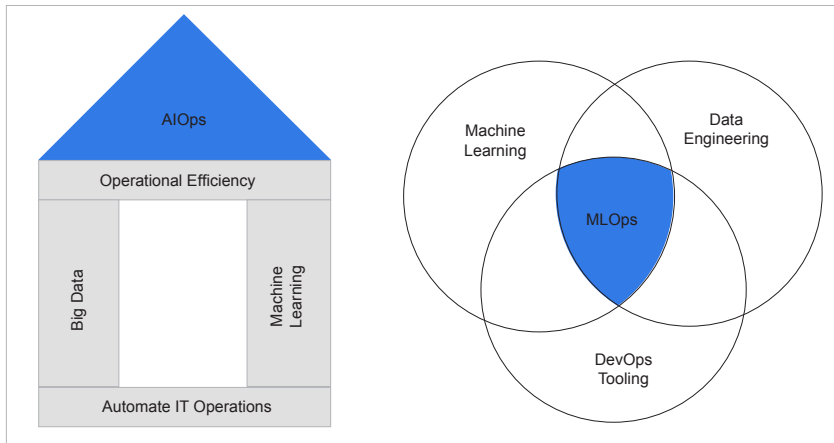


Figure 2: AIOps versus MLOps

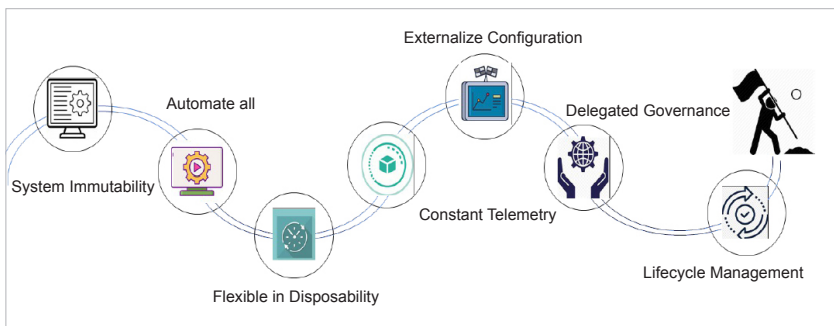


Figure 3: Key principles of cloud management tools

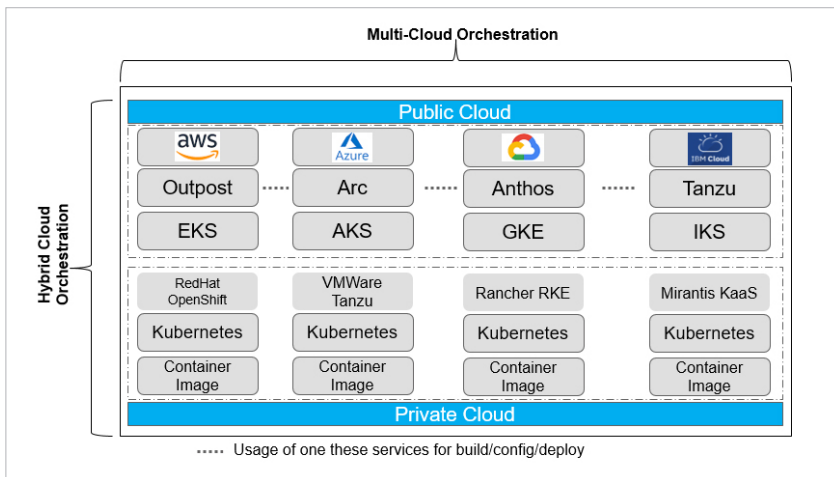


Figure 4: Multi-cloud management tools that use cloud agnostic services

automatically, the toolchain process in cloud provisioning and DevOps activities can be improved.

3. Improved customer satisfaction with less cloud adoption cost:

Automating operational activities like data driven insights, monitoring

and dashboard solutions, and action for events based on knowledge based policies and rule guidelines, helps in building overall customer satisfaction with minimal or no human intervention.

AIOps focuses on three services:

data, insights and actions. Data provides the input for the design, which includes historical behaviour of problems and the solution that was carried out (e.g., shutdown instances when resource usage of a virtual machine is critically high). Insights give the detailed trained model for a given problem and bisect the data to detect it, diagnose the solution and optimise the search metadata tags. Actions define the proposed command that needs execution based on data input and insights models. This service defines how to mitigate a problematic situation, integrate it with ticketing solutions like ServiceNow, optimise cloud services based on best practices, and improve process efficiency in operational activities like infrastructure provisioning, reporting and alert services.

By using AIOps many cloud service providers are improving incident management. They integrate their service with cloud operational activities like incident reporting, triage for priority handling, mitigation/resolution, anomaly detection in end-to-end monitoring of infrastructure, along with actions based on identified issues such as system failures and high resource usage. For a safe cloud deployment and for improved efficiency, there are a few additional things that need to be looked into such as the smoke/regression test, static review of code, configuration, and health checks of the environment.

In short, the movement of DevOps to CloudOps to improve native operational activities can be done with machine learning and automation solutions using AIOps.

Cloud management tools at a glance

Cloud management tools play a vital role in improving productivity and simplifying operational activities including cost management, resource management and governance. Some of the key parameters that should be looked at while choosing these tools are listed in Figure 3.